Wheel Brakes Unit (Mod 6/2002) – General Data

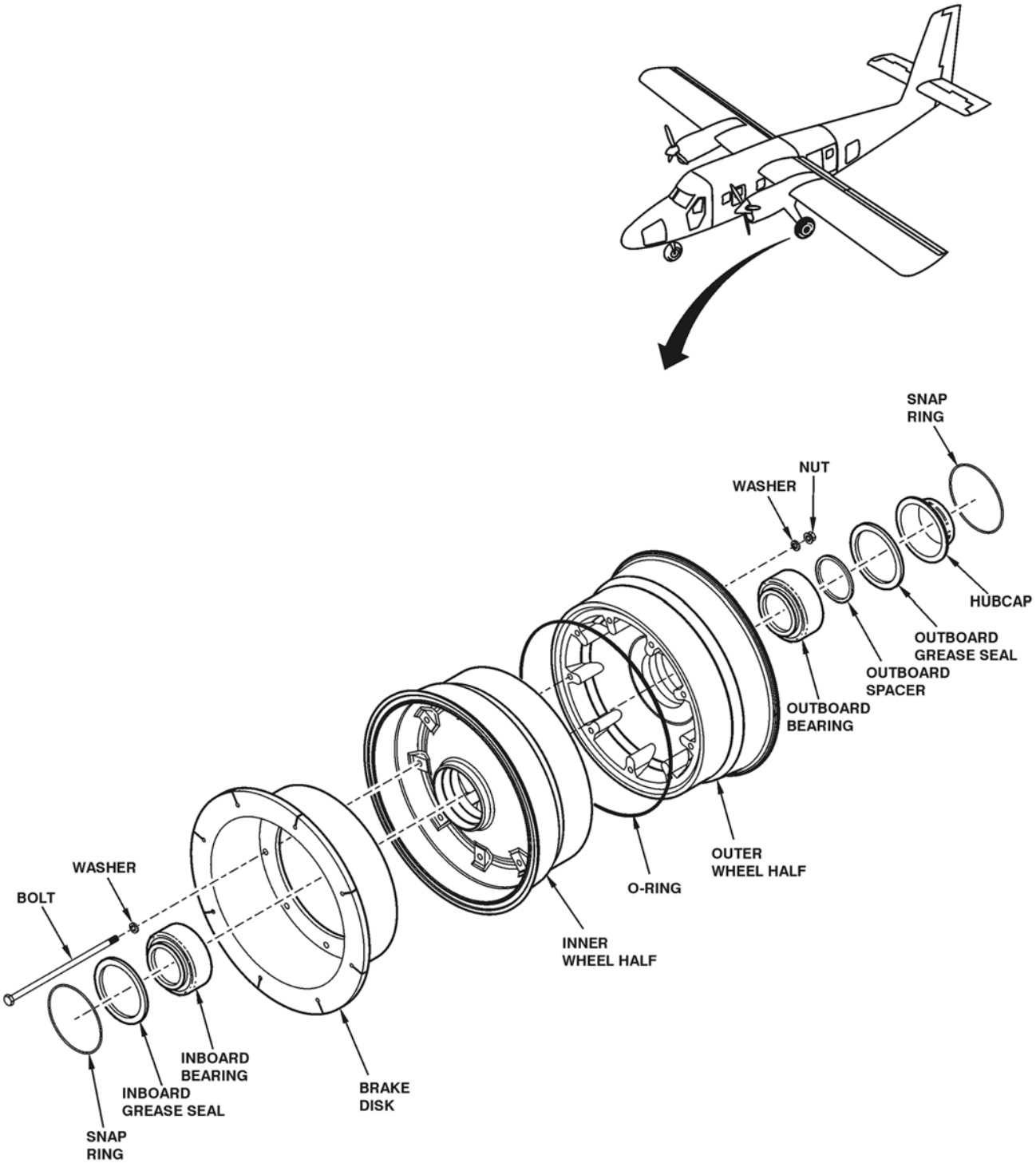
1. General

A wheel and brake unit is bolted to the torque plate on each mainwheel axle. The brake assembly is a piston actuated, hydraulically operated, external disc unit designed for use with a petroleum base hydraulic fluid. The right and left brake assemblies can be interchanged by switching the bleeder fitting with an inlet fitting.

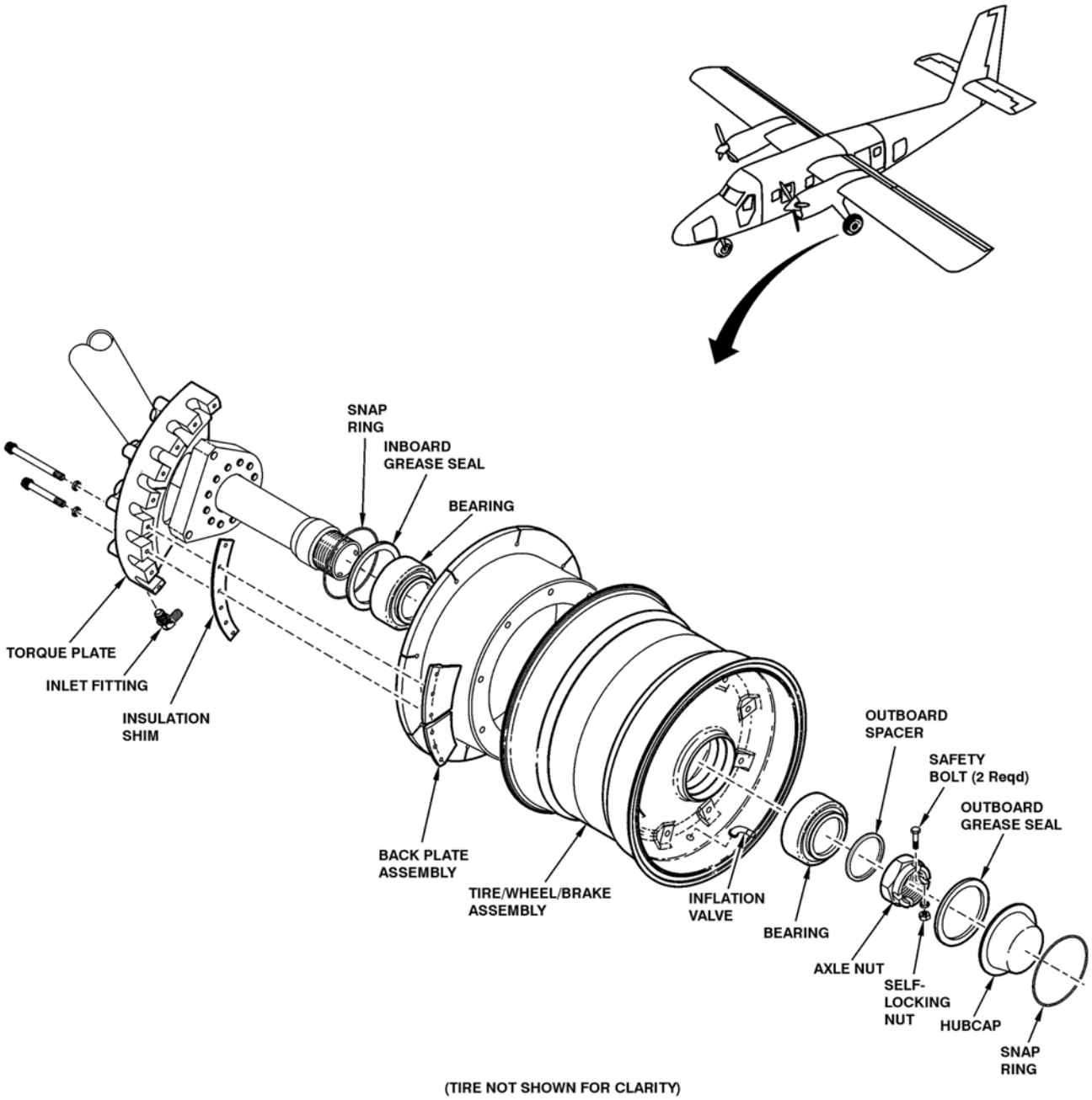
Each brake assembly is composed of a pressure plate assembly, four pistons, four backing plate assemblies, a torque plate, a cast aluminum alloy cylinder housing, four piston insulators to minimize the conduction of heat into the hydraulic fluid, four O-rings, four piston boots to seal out contaminants, four friction springs to maintain ideal piston positioning, two sets of anchor bolt hardware, a hydraulic fitting, one set of bleeder fitting hardware, and twelve high strength bolts, nuts, and washers to fasten the four back plate assemblies to the cylinder housing.

Braking action occurs when hydraulic pressure is applied to the brake via the pilot's or co-pilot's brake control valves. As hydraulic pressure reaches the caliper, the pistons move outward in the caliper forcing the pressure plate against the surface of the brake disc. This causes the caliper to align itself so that the pressure plate linings and backplate linings provide uniform contact against both rubbed surfaces of the brake disc.

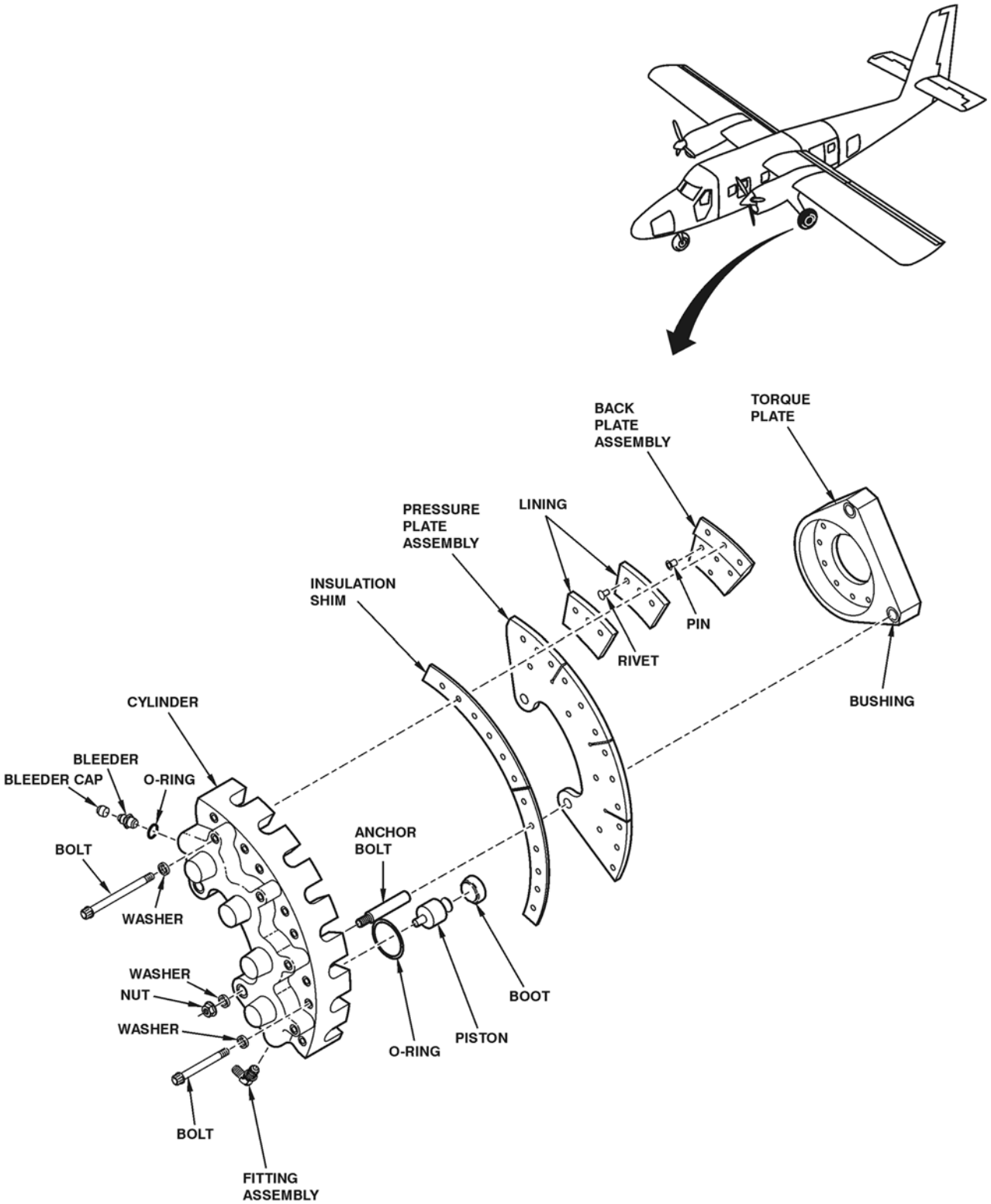
Refer to manufacturers information regarding maintenance, servicing repair of the brake assembly.



Wheel and Brake Assembly
Figure 1 (Sheet 1 of 3)



Wheel and Brake Assembly
Figure 1 (Sheet 2 of 3)



Wheel and Brake Assembly
Figure 1 (Sheet 3 of 3)

2. Removal/Installation

A. Brake Unit Removal

1) Discharge brake accumulator by pumping brake pedals.

2) Remove and cap brake hydraulic line.

3) Remove the twelve brake cylinder tie bolts and washers, and four back plate assemblies. (Refer to Figure 1 Sheet 3 of 3).

4) Slide the cylinder out of the torque plate. (The torque plate will remain mounted to the axle).

5) Place the assembly on a clean flat surface.

6) Remove the pressure plate assembly.

B. Brake Unit Installation

1) Install the pressure plate assembly (the lining side must NOT face the pistons) by aligning the bolt holes with the anchor bolts and slide the pressure plate onto the cylinder. The pressure plate MUST float freely on the anchor bolts.

2) Install the brake housing into torque plate by aligning the anchor bolts with the torque plate holes and sliding the brake assembly onto the torque plate. It MUST slide freely.

3) Place a countersunk washer on each of the twelve brake tie bolts, and insert these bolts into the respective brake housing holes.

Note The countersunk side of the washer must face the bolt head or bolt failure may result.

4) Install two new insulator shims onto brake housing.

5) Install the four backplate assemblies onto the brake housing and lubtork (MIL–T–5544A) the brake tie bolts to 80 – 85 in–lbs.

Note Be sure that the brake disc is sandwiched between the backplate and the pressure plate.

6) Reconnect the hydraulic lines.

7) Bleed brake with procedure from 32-40-00, Wheels and Brakes (Mod 6/2002) – General Data.

8) Check the pedal for proper feel and travel.

Note Wheels should rotate freely. There should be no evidence of binding or excessive brake drag.

3. Inspection/Check

Do inspections in accordance with manufacturer's current maintenance instructions, detail check.

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